Ashwath Balaji

Open to Relocation • 804-418-9085 • ashwathbalaji04@gmail.com • linkedin.com/in/ashwathb/ • github.com/AshwathB-debug

Education

Virginia Commonwealth University Graduated: May 2026
B.S. in Computer Science, Concentration in Data Science, Minor in General Business GPA: 3.4
Coursework: Algorithm Analysis with Advanced Data Structures, Machine Learning, Intro to Natural Language Processing

Experience

Associate Consultant, Ignyte Group Jun 2026 - Aug 2026

- **Mapped an end-to-end licensing workflow** (application submission, automated case routing, and agent/manager reviews).
- **Authored** and **prioritized 20 user stories** using **MoSCoW** methodology **translating complex processes** into structured tasks.
- **Prototyped** a **ServiceNow** duplicate-case detection feature using **all-MiniLM** model from Hugging Face to capture **semantic similarity** on short-description text to flag likely duplicate submissions.

Product Engineer Intern, Nestease Dec 2025 - Jan 2026

- **Built** a **Python/Gemini API PoC** that automates image analysis to calculate measurements and assign base prices for target objects from an Excel sheet; proposed a caching strategy to store repeat use input tokens projected to cut API costs **~48%.**
- **Led end-to-end definition** for an **AI pricing feature**, **documenting 14** functional specifications in a **PRD** with detailed **acceptance criteria**, **user stories**, **Figma wireframes**, and **MoSCoW prioritization**. **Demoed functionality to leadership**, driving roadmap adoption.
- **Wrote** a **zero-shot Chain-of-Thought** system prompt that aimed to reduce hallucination on a handful of edge case images in measuring target objects and assigning base prices, prioritizing accuracy over latency.

Software Engineer Intern, Kolabrata.au Jun 2025 - Nov 2025

- **Developed** a stateless **RESTful Flask** backend, implementing secure **REST API endpoints** and JSON serialization to facilitate frontend-backend communication. Collaborated with a **cross-functional team** to optimize Kolabrata's website.
- Refactored monolithic script into a **modular Object-Oriented architecture,** implementing **Service-Controller patterns** to decouple business logic from Gemini API integration, **reducing script complexity by ~60%** in lines of code**.**

Undergraduate Researcher, VCU College of Engineering May 2025 - Sep 2025

- **Built** a Flutter/Dart **iOS** interface with camera controls and guided overlays to fix region-of-interest (ROI) misalignment.
- **Developed** a Python/OpenCV algorithm that closely approximated midbrain-to-pons ratios on test MRI images and compared output against manually measured ratios; connected backend-frontend components through a Flask REST API.

Projects

AI Agent for Scientific Visualization, VCU College of Engineering (Capstone Project) Aug 2025 - Oct 2026

- **Led a 4-person team** to build a **multi-agent** workflow on **Qwen 3** (Hugging Face) that analyzes datasets via **multi-step prompts** and generate **Matplotlib** charts. Led a **Streamlit** UI integration and deployed app to **AWS Fargate (ECS).**
- **Led weekly Agile standups** and presented progress and research findings to stakeholders. **Gathered requirements** and mapped **end-to-end** process diagrams in draw.io translating stakeholder insights into agentic solutions.

Cricket RAG System, Natural Language Processing Apr 2026 - Oct 2026

- **Built an end-to-end RAG application** in Python using **Qwen 3** (Hugging Face) grounded on Cricket data from Wikipedia. **Evaluated** a **50-question** test set using **Claude API** (LLM-as-judge), improving average answer accuracy from **~70%** to **82%** and retrieval hit rate of **86**% by tuning model settings (temperature, repetition penalty, thinking) and few-shot examples.
- **Designed** an **ETL** pipeline using **LlamaIndex** that dynamically extracts Wikipedia data, chunks sections, strips extraneous content, and embeds data in **Pinecone** vector database; integrated a **Streamlit** chat interface for querying.

Financial Data Manager, Databases Oct 2024 - Dec 2024

- **Designed a normalized MySQL database** for the Magnificent 7 stocks. Ingested financial data from Yahoo Finance/Alpha Vantage. Developed **SQL** code in Python and **SQLAlchemy**, retrieving essential data to support analysis.

Skills

AI/Integration: Prompt Engineering, Hugging Face, Gemini API, Claude API, RAG, LlamaIndex, ETL, REST API (POST)
Languages: Python, SQL, Java, C, Dart
Operations/Methodologies: Agile (Scrum), Requirements Gathering, Technical Documentation, Workflow Automation, Process Mapping (draw.io), PRD, Prioritization (MoSCoW)
Databases/Frameworks/Libraries: MySQL, Pinecone, Streamlit, Pandas, NumPy, Flask, OpenCV, Flutter
Tools: Git, Linux, Bash, Microsoft Excel, ServiceNow (Customer Service Management)

Certifications

ServiceNow: Certified System Administrator (2026), Certified Implementation Specialist: Customer Service Management (2026)